Smit Parewa

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.52
KENDRIYE VIDALAYA NO 3 JAIPUR	Class XII (CENTRAL BOARD SECENDORY EDUCATION)	2019	68.00%
KENDRIYE VIDALAYA NO 3 JAIPUR	Class X (CENTRAL BOARD SECENDORY EDUCATION)	2017	10.00%

POSITIONS OF RESPONSIBILITY

- Teaching Assistant in Machine Learning Course | July to November 2025.
- Teaching Assistant in Chemical Reaction Engineering-1 course | Jan-May 2026.
- Teaching Assistant in Computational Biology | July - Nov 2026.
- Teaching Assistant in Cardiovascular Mechanics | July-Nov 2026.

WORK EXPERIENCE

Plant Engineering Intern | Wave Distilleries & Breweries Ltd. *(March 2024 - June 2024)*

- Process Data Tracking: Checked daily temperature, flow rates, and yield data to improve plant work and reduce mistakes.
- Problem Solving: Analyzed production data to find reasons for delays and help fix batch problems.
- Daily Reporting: Prepared clear daily summaries from factory data to help managers make better decisions.
- link https://drive.google.com/drive/folders/1zBXPw7nuuiEoCNeKx3zEB2OxYGUf74p-?usp=sharing

PROJECTS

Thesis Project (MTech)-Arterial Stenosis Geometry Tracing and Parameter Estimation Using Deep CNNs and OpenCV Color Vision Techniques

Mentor: Dr Anand Mohan, Chemical Engineering, IIT Hyderabad

- ongoing
- Custom Medical Dataset Creation: Extracted 30 Gaussian mixture parameters (a, mu, sigma) into a CSV file and paired them with cropped coronary Angiogram images to create a labeled training dataset for plaque geometry.
- Deep CNN Regression Model: Trained a ResNet18 neural network in PyTorch to directly predict all 10 Gaussian parameters from blood vessel images to trace stenosis severity.
- Model Optimization & Training: Used transfer learning, target scaling, 5-Fold Cross-Validation, and model ensembling to prevent overfitting on small medical datasets.
- Automated Curve Overlay: Built a Python script using OpenCV and Matplotlib to reconstruct 10-Gaussian curves and overlay them directly onto vascular images for visual verification.

Nanofluid Thermal Conductivity Prediction using Artificial Neural Networks (ANN)

- Predictive Modeling: Developed an Artificial Neural Network (ANN) to predict nanofluid thermal conductivity from experimental datasets.
- Data Preprocessing: Handled missing values, outliers, and feature normalization to prepare complex thermophysical data for training.
- Model Optimization: Tuned network architecture, activation functions, and hyper-parameters to minimize prediction error and prevent overfitting.
- Performance Evaluation: Evaluated ANN performance using key regression metrics, including MSE, RMSE, and R^2 score.
- Git Hub Link : https://github.com/smitparewa123-cmd/Nanofluid-Thermal-Conductivity-Prediction-using-Artificial-Neural-Networks-A

NN-

RAG-based AI Teaching Assistant (Gen AI & LLMs)

- Built a smart AI chatbot by integrating Large Language Models (LLMs) and vector databases.
- Connected the AI to course notes and videos so it answers student questions accurately.
- Created a web interface for real-time streaming answers without AI hallucinations.
- Git Hub Link: https://github.com/smitparewa123-cmd/RAG-based-AI

Real Estate Price Prediction & Market Analysis

- Balanced Data Splitting: Separated the housing data based on income levels so the training set fairly represents all income groups instead of picking data randomly.
- Automated Data Cleaning: Built a pipeline using the scikit-learn library to automatically fill missing values, standardise numbers, and convert text categories into numbers.
- Model Testing & Comparison: Trained Linear Regression, Decision Tree, and Random Forest models, then tested their accuracy using 10-fold cross-validation to find the best predictor.
- Git Hub Link: https://github.com/smitparewa123-cmd/Real-Estate-Price-Prediction-Market-Analysis

Instagram User Data Analysis & Scraping

- Raw Data Parsing: Cleaned raw text files into structured JSON data using custom Python code.
- Smart Data Conversion: Handled missing information and converted shortened numbers (like K and M) into full digits.
- Profile Metrics Analysis: Analyzed follower stats, post counts, and grouped users into 34 different categories.
- Git Hub Link: https://github.com/smitparewa123-cmd/Instagram-User-Data-Analysis-Scraping

Social Network Page Recommendation System

- User Data Processing: Loaded and processed large-scale JSON user interaction data using Python set operations.
- Recommendation Algorithm: Built a collaborative filtering algorithm to suggest pages based on shared user interests.
- Score-Based Ranking: Calculated page affinity scores and ranked top recommendations for target users.
- Git Hub Link: https://github.com/smitparewa123-cmd/Social_network_page_recomandation_system

SKILLS

- Python, SQL,Data Cleaning & Preprocessing, Exploratory Data Analysis (EDA)
- Pandas, NumPy, Matplotlib, Seaborn,Practical Machine Learning, Scikit-learn
- Probability & Statistics, Probability Distributions,Retrieval-Augmented Generation (RAG), Large Language Models (LLMs)
- Vector Databases, Neural Networks (Deep Learning)
- Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Conda
- Web Development for Data Science

SCHOLASTIC ACHEIVEMENTS

- Certificate of completion of the Data Science Course by Code With Harry.
- Certificate of completion of the Python Course by Code With Harry.
- Secured 83.36 % in TCS ION NQT | JULY 2024.
- Secured 2nd Position in a quiz competition organized by the Department of Chemical Engineering at IIT Hyderabad.
- Link https://drive.google.com/drive/folders/1CTfZcyLvagpxr4FJ64s8aoFY0uYiX752?usp=sharing

EXTRACURRICULAR

- Active Core Member of the Speaking Art and Expression Club Rang De Manch (RDM), IIT H.
- Active member of the Fashion Club Silhouettes, IIT H.
- Active Member of the Filmmaking Club Behind The Lens (BTL) at IIT H.